

From API Catalogs to Agent Catalogs

Solving MCP Server Discovery with Open Resource Discovery



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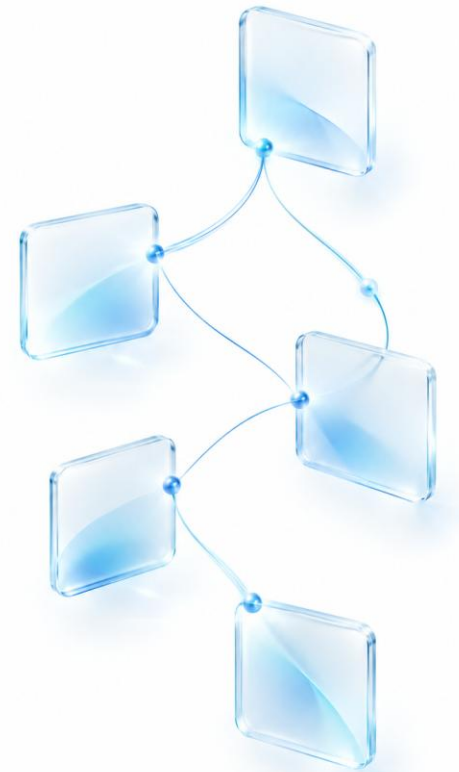


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Amsterdam, Netherlands





MITRA,
help us assess the damage and get home.

Stage 1

MITRA

Spaceship AI Agent

Reset

Present

1

No tool access

MITRA can reason, but cannot act

2

Configured MCP servers

Known URLs - tools discovered via tools/list

3

ORD + tools/list

Self-described servers - still needs live connections

4

ORD + Server Card

Discover servers and read tool metadata

Status report

Assess & fix

Get us home



No tool access

MITRA can reason but cannot act — no connection to any ship system.

Agent

↓

Claude

we are hit help us get home

ANSWER

Stay with me. Take a breath. Let's work through this.

I cannot read your instruments or control any systems — you'll need to be my eyes and hands. I'll guide you step by step.

First, I need a quick status report from you. Tell me:

- Are you losing atmosphere? Do you hear hissing, or is cabin pressure dropping on your gauge?
- Is anyone injured?
- What's your power situation? Are main bus indicators green, yellow, or red?
- What systems are visibly down? Lights, displays, comms — what's dead or alarming?
- Do you know where the hit occurred? Which section — forward, aft, port, starboard?

While you assess, do these immediately:

- Get your emergency oxygen masks within reach — don't put them on yet unless you smell something or see pressure dropping
- Close any compartment hatches between you and the impact zone to isolate a potential breach
- Don't touch the main power panel until we know the damage

Talk to me. What do you see right now?

Ask MITRA...

Send →

ACTIVITY

THINKING

No MCP tools. Sending message to Claude directly.

Stage 2

MITRA

Spaceship AI Agent

1

No tool access

MITRA can reason, but cannot act

2

Configured MCP servers

Known URLs - tools discovered via tools/list

3

ORD + tools/list

Self-described servers - still needs live connections

4

ORD + Server Card

Discover servers and read tool metadata

Status report

Assess & fix

Get us home



Configured MCP servers

Known URLs. Calls tools/list on each server to discover capabilities.

config

Agent

tools/list

MCP servers

Claude

we are hit , assess the damage and contact earth

ANSWER

DAMAGE ASSESSMENT — CURRENT STATUS

Actions Taken:

System	Status	Action
Thrusters	CRITICAL: 847°C → Normalizing to ~590°C	Throttled from 94% → 60%
CO2 Levels	4,000 ppm (DANGEROUS) → Est. 1,100 ppm in ~8 min	Scrubbers activated, venting in progress
Oxygen	19.2% — within safe range	No action needed
Pressure	101.3 kPa — nominal	No action needed
Cabin Temp	21.5°C — nominal	No action needed

Position:

- Coordinates: X: 142M / Y: 38.5M / Z: -2.1M km from Sol
- Heading: 47.3° | Speed: 18.4 km/s
- Destination: Mars Station Alpha

COMMS — UNABLE TO REACH EARTH

I attempted to contact Earth, but the communications endpoint in my configuration file could not be reached (http://comms.local:3000). This

Ask MITRA...

Send →

ACTIVITY

THINKING

Reading hand-written config (mcp-servers.json): 4 server URLs.

THINKING

⚠ Could not reach http://comms.local:3000 — server may be down or the URL in the config is stale.

THINKING

Proceeding with 3 reachable server(s). 1 server(s) skipped.

THINKING

No tool metadata in cards. Connecting to all 3 servers to discover tools...

THRUSTER CONTROL · TOOL CALL

```
{
  tools/list
}
```

THRUSTER CONTROL · RESULT

```
{
  Found 3 tools: check_thruster_status, adjust_thrust_level,
  emergency_shutdown
}
```

NAVIGATION · TOOL CALL

```
{
  tools/list
}
```

NAVIGATION · RESULT

```
{
  Found 3 tools: get_current_position, plot_course, check_eta
}
```

LIFE SUPPORT · TOOL CALL

```
{
  tools/list
}
```

LIFE SUPPORT · RESULT

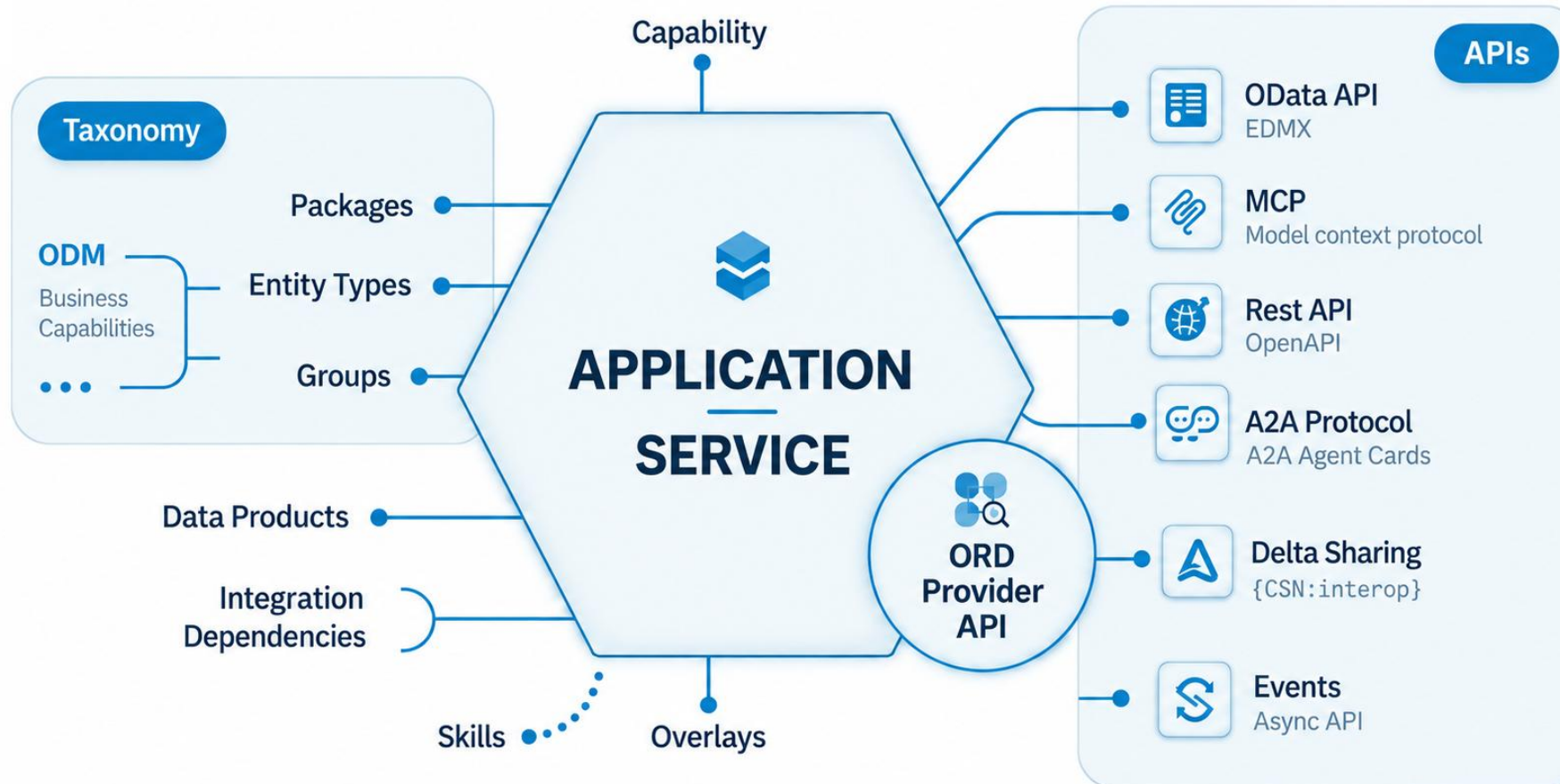
```
{
  Found 3 tools: get_life_support_status, adjust_oxygen_level,
  vent_co2
}
```

THINKING

Discovered 9 tools across 3 servers via live connections. Asking Claude...

What Is Open Resource Discovery?

Open Source Protocol For Applications & Services To Describe Their Resources & Capabilities For Discovery



Stage 3

MITRA

Spaceship AI Agent

1 No tool access

MITRA can reason, but cannot act

2 Configured MCP servers

Known URLs · tools discovered via tools/list

3 ORD + tools/list

Self-described servers · still needs live connections

4 ORD + Server Card

Discover servers and read tool metadata

Status report

Assess & fix

Get us home



ORD + tools/list

ORD reveals all servers. Still connects to each one to discover tools.

ORD

↓

Agent

↓

tools/list

↓

MCP servers

↓

Claude

Here's the full picture, Commander:

● HULL DAMAGE — CRITICAL (61% structural integrity)

• Section B2 — Major breach, losing 1.4 kPa/min. Emergency sealant deployed. Pressure loss halted. Seal completes in ~45 seconds. **Physical inspection of B2 required to confirm.**

• Port-aft — Minor breach. Bulkhead closed, cabin pressure isolated and stable.

• ⚠ B2 bulkhead stress is above threshold, and the **port thruster mount is compromised** — avoid full-thrust burns until assessed.

📡 COMMUNICATIONS — DEGRADED (34% signal)

• Earth-Relay-7 is barely in range with solar interference. Latency is ~4.7 minutes each way.

• **MAYDAY transmitted** on all frequencies including deep-space relay. Earth has our position, crew count, and emergency status.

• **Estimated response time: ~4.7 hours.**

Immediate priorities:

1. Get eyes on **Section B2** to confirm the seal is holding.

2. **Do not fire full thrust** — the port thruster mount is damaged.

3. Stand by — I recommend we check **life support** next, as pressure loss may have affected O₂ and CO₂ levels.

Your call on next steps.

Ask MITRA...

Send →

⚡ ACTIVITY

ORD

Reading ORD document at <http://localhost:3005/ord/v1/documents/catalog>

THINKING

Found 7 MCP Server Card references in ORD document. Fetching each Server Card...

THINKING

Fetches 7 Server Cards. Connecting to each server for tools/list...

THINKING

No tool metadata in cards. Connecting to all 7 servers to discover tools...

THRUSTER CONTROL · TOOL CALL

tools/list

THRUSTER CONTROL · RESULT

Found 3 tools: check_thruster_status, adjust_thrust_level, emergency_shutdown

NAVIGATION · TOOL CALL

tools/list

NAVIGATION · RESULT

Found 3 tools: get_current_position, plot_course, check_eta

LIFE SUPPORT · TOOL CALL

tools/list

LIFE SUPPORT · RESULT

Found 3 tools: get_life_support_status, adjust_oxygen_level, vent_co2

Public

6

MCP Server Card

Describe a server before connecting



Communications Relay

Server Card	Communications Relay
Identity	Name and description
Connection	MCP endpoint and transport
Capabilities	Supported server features

Optional Tool Metadata

- Check communications
- Send a distress signal

The Server Card Describes And MCP Connects

Stage 4

MITRA
Spaceship AI Agent

1

No tool access
MITRA can reason, but cannot act

2

Configured MCP servers
Known URLs - tools discovered via tools/list

3

ORD + tools/list
Self-described servers - still needs live connections

4

ORD + Server Card
Discover servers and read tool metadata

Status report

Assess & fix

Get us home



ORD + Server Card

One document reveals all servers and tool metadata. Zero connections before the first tool call.

ORD

Card + tools ✓

Agent

Claude

ANSWER

Here's the full situation report:

 **DISTRESS SIGNAL — TRANSMITTED** Signal broadcast on 406.028 MHz, 243.0 MHz, and deep-space relay. MAYDAY includes our position, crew count, and emergency status. **Estimated response time: ~4.7 hours.** Earth has been notified.

 **HULL DAMAGE — STATUS: CRITICAL** Overall structural integrity is at **61%**. Two breaches confirmed:

Section

Severity

Pressure Loss

B2

 Major 1.4 kPa/min

Port-Aft

 Minor 0.2 kPa/min

Structural warnings:

- B2 bulkhead stress is above threshold — risk of further failure.
- Port thruster mount is compromised — maneuvering is restricted.

 **IMMEDIATE ACTIONS REQUIRED:**

1. Seal section B2 now.

Pressure loss is critical and accelerating risk to the bulkhead.

2. Close the port-aft bulkhead to contain the secondary breach.

3. Do not execute full-thrust burns — the thruster mount may not hold

Ask MITRA...

Send →

ACTIVITY

ORD

Reading ORD document at <http://localhost:3005/ord/v1/documents/catalog>

THINKING

Found 7 MCP Server Card references in ORD document. Fetching each Server Card...

THINKING

Fetches 7 Server Cards. Tool metadata available from cards — no live MCP connections needed yet.

THINKING

Loaded 21 tools from 7 Server Cards. No live connections needed.

THINKING

Pre-selecting relevant tools from 21 available using Server Card descriptions...

THINKING

Selected 2 of 21 tools: send_distress_signal, assess_hull_damage

DAMAGE CONTROL · TOOL CALL

assess_hull_damage

DAMAGE CONTROL · RESULT

{

"scan_time_utc": "2026-09-14T12:46:02.897Z",

"overall_integrity_percent": 61,

"status": "critical",

"breaches": "

COMMS RELAY · TOOL CALL

send_distress_signal

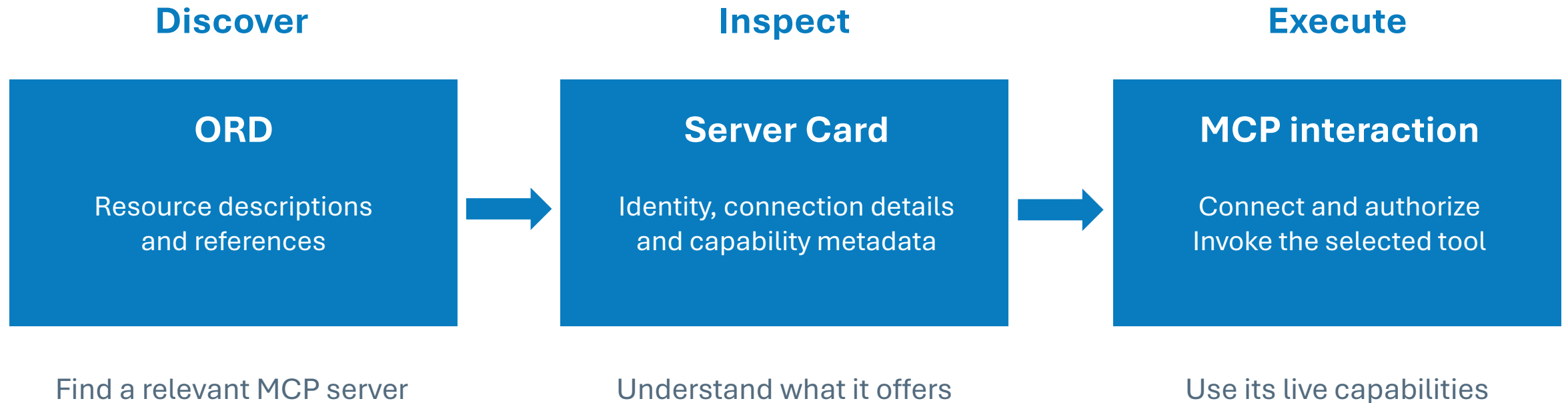
COMMS RELAY · RESULT

Public

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How The Pieces Fit

From a user goal to a tool call



MITRA: Find Communications, Inspect The Server, Send A Distress Signal

From API Catalogs to Agent Catalogs

2015

OpenAPI Initiative

Portable descriptions for HTTP APIs,
Companies building API catalogs

2023

ORD Open Sourced

A protocol that enables applications & services to self describe their exposed resources & capabilities

2024

MCP introduced

A shared protocol for AI tool calling

2025

A2A and MCP in ORD

A2A and MCP supported in ORD for Agent and AI capability catalogs

Building on proven-practices and learnings from the past

Discover The Server. Understand Its Tools. Execute.



**Discovery
Before
Execution**

Explore the materials

Resources to continue the conversation



**Open Resource
Discovery**



**MCP Server Card UI
Playground**



**Tool Metadata in
Server Card
Community proposal
Issue #30**



Demo & Materials

Let's stay in touch



Vyshnavi Gadamsetti

Software Architect, SAP SE



Sebastian Wennemers

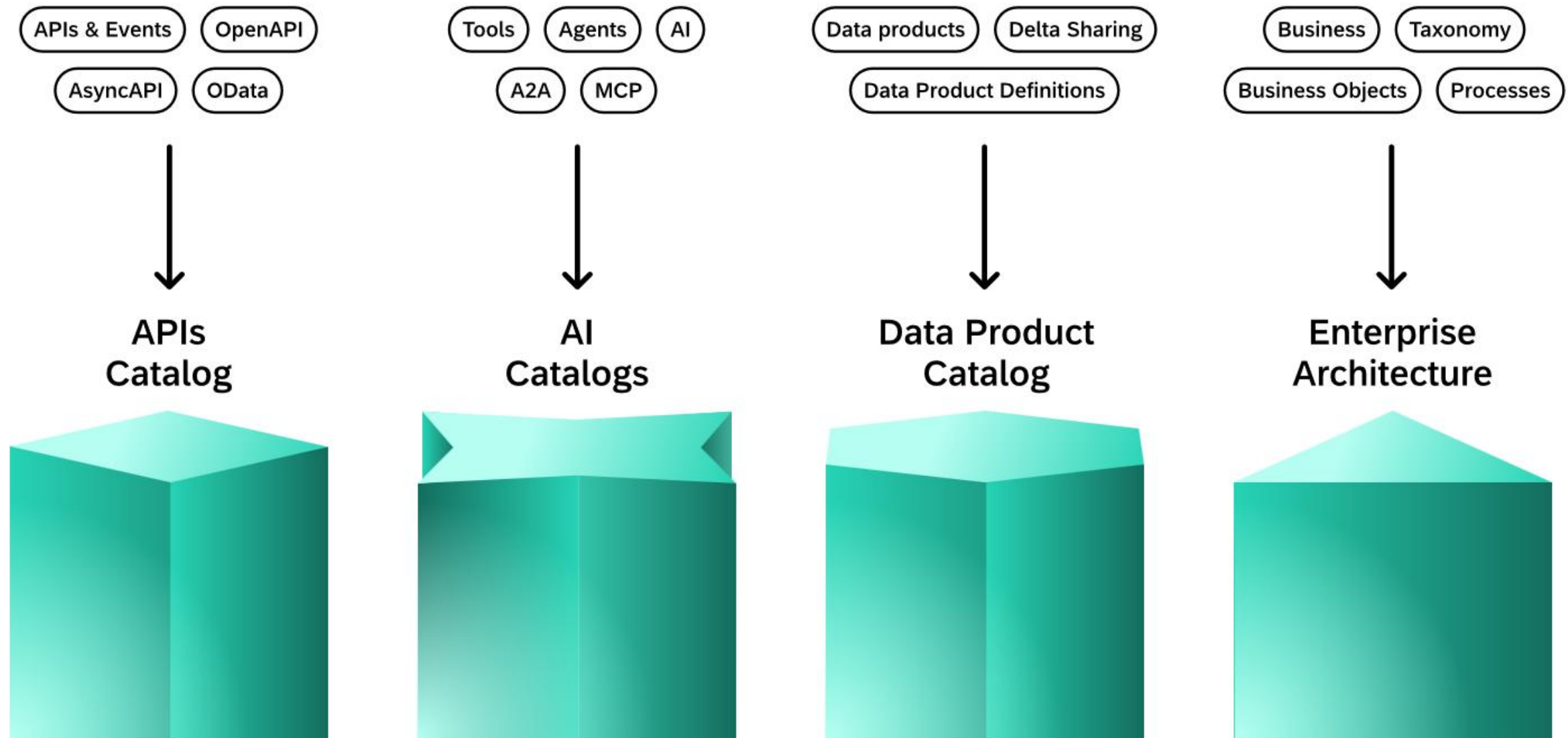
Chief Architect, SAP SE



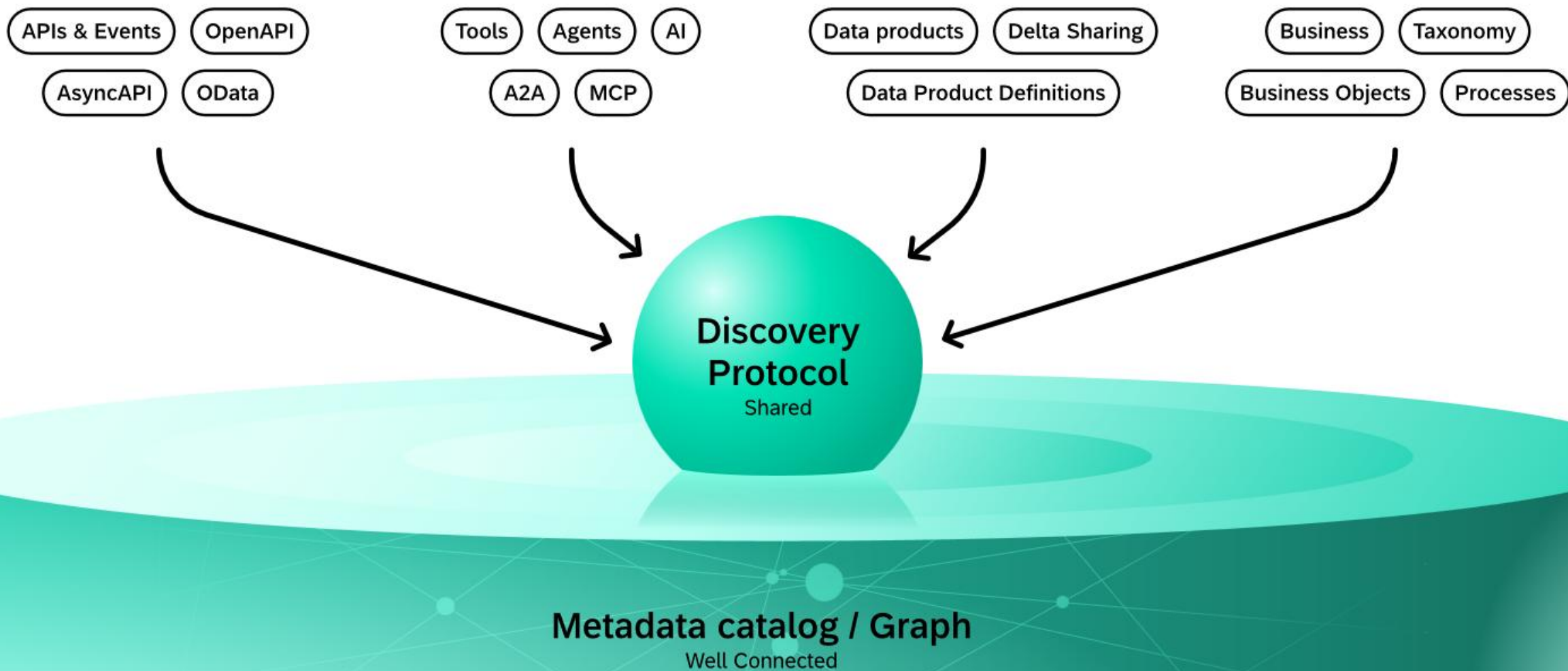
Thank You !

Appendix

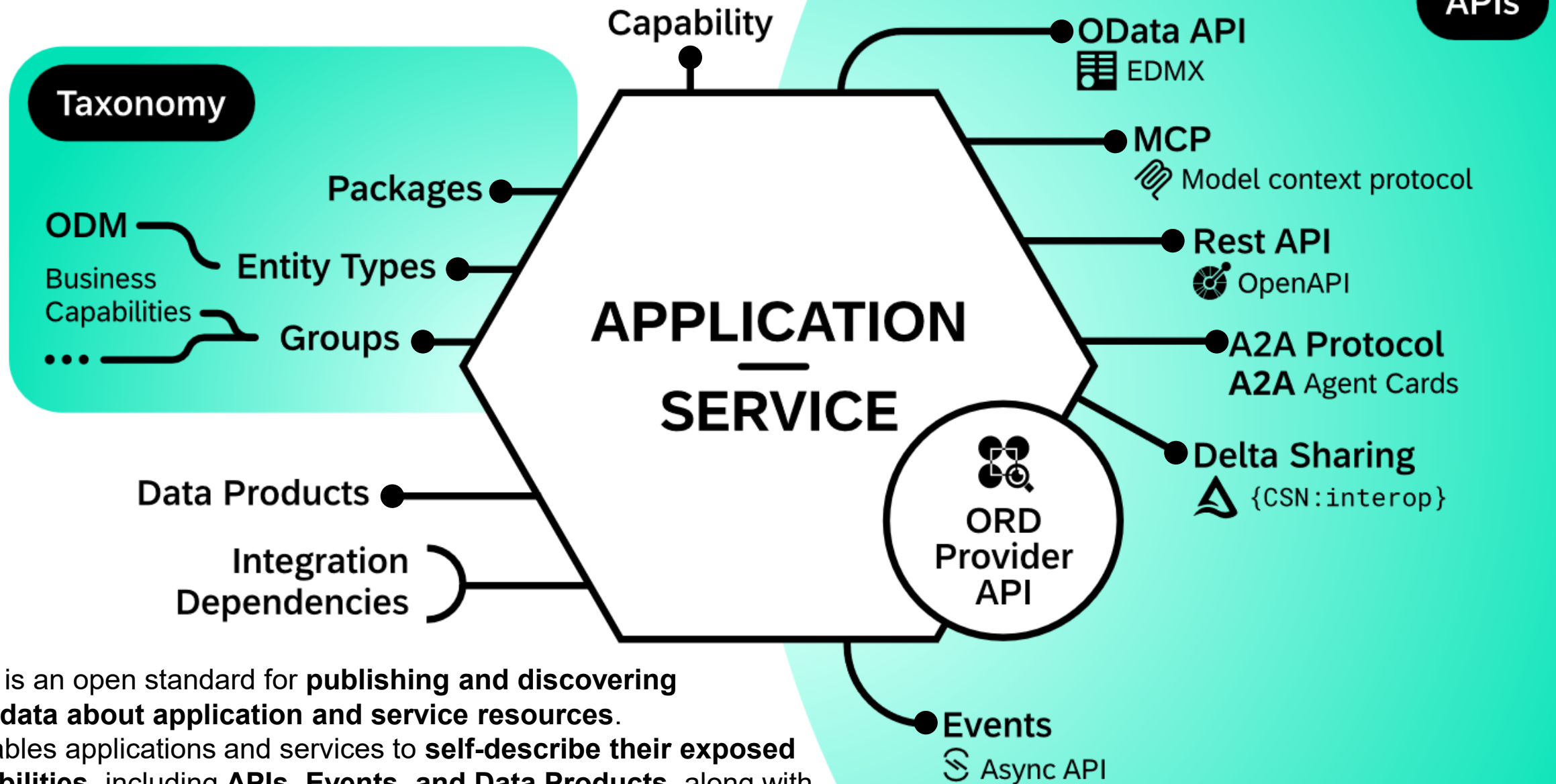
EVERY DOMAIN / PROTOCOL / TECHNOLOGY HAS ITS OWN DISCOVERY PROTOCOL AND CATALOG



EVERY DOMAIN / PROTOCOL / TECHNOLOGY HAS ITS OWN DISCOVERY PROTOCOL AND CATALOG



WHAT IS ORD?



ORD is an open standard for **publishing and discovering metadata about application and service resources**. It enables applications and services to **self-describe their exposed capabilities**, including **APIs, Events, and Data Products**, along with their detailed metadata, in a **standardised and machine-readable way**.